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BAGONG PILIPINAS

CC 101 – Introduction to Computing

"From Problem to README": Designing a Number-System Converter and Documenting It in Markdown

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Computer Unit Used (Desktop/Laptop, brand): HP **Date: October 20, 2026**

I. Objectives

At the end of this activity, you should be able to:

- Analyze a number-system conversion problem by identifying its inputs, processes, loop condition, and outputs.
- Write pseudocode using sequence, selection, and iteration.
- Draw flowcharts using standard symbols and check them by tracing and testing with sample data.
- Create a properly formatted README.md file in Markdown using Notepad, without an Internet connection.
- Upload the README.md to your CC 101 GitHub Computing Portfolio repository (take-home part).

II. Materials Needed

- A computer running Windows 11 with Notepad (no network or Internet connection is needed in the laboratory)
- This worksheet, a pencil, and a ruler (or a flowchart template) for drawing flowcharts
- A USB flash drive, to bring your README.md home for uploading
- For Part G only (outside the laboratory): any device with Internet access and a free GitHub account
- Week 7 lecture notes, for reference on pseudocode, flowchart symbols, and Markdown syntax

III. Background

Programmers do not start by typing code. They first analyze the problem, plan the solution as pseudocode and a flowchart, and check it by hand. In this activity, you will design a program that converts numbers between the decimal and binary number systems you learned in Week 5. An algorithm can be written as pseudocode (English-like steps) or drawn as a flowchart (standard symbols connected by arrows). Good work is also documented. On GitHub, the standard way to introduce a project is a README.md file written in Markdown, a plain-text format that uses symbols such as #, *, and - for headings, emphasis, and lists. Because Markdown is plain text, you can write it in Notepad with no Internet connection and upload it to GitHub later.

IV. Procedure

Part A – Analyze the Problem (Problem 1)

Problem 1: Decimal-to-Binary Converter. Design a program that asks the user for a whole number in **decimal** (base 10) that is 0 or greater, converts it to **binary** (base 2), and displays the binary

result.

Worked example (13 in decimal): 13 ÷ 2 = 6 r 1; 6 ÷ 2 = 3 r 0; 3 ÷ 2 = 1 r 1; 1 ÷ 2 = 0 r 1. Reading the remainders from bottom to top gives **1101**.

1. Convert 13 and 25 to binary by hand using the method above, so you understand every step before writing the pseudocode.
2. Complete **Table 1**: list the input, the output, the processes (operations), the decision or loop condition, and the variables you will need.

Part B – Write the Pseudocode

3. In **Box 1**, write the pseudocode for Problem 1. Use START, INPUT, SET, WHILE … DO … ENDWHILE, IF … THEN … ELSE … ENDIF, DISPLAY, and END, and indent the steps inside the loop.

   Tip: Use **MOD** for the remainder of a division (13 MOD 2 = 1) and **DIV** for whole-number division (13 DIV 2 = 6). Each new remainder must be placed at the **left** of the binary digits found so far, for example SET binary = remainder + binary. Also decide what your algorithm should display when the input is 0.

Part C – Draw the Flowchart and Desk Check

4. In **Figure 1**, draw the flowchart for Problem 1 using the standard symbols: oval (start/end), parallelogram (input/output), rectangle (process), and diamond (decision). Show the loop with an arrow going back to the decision, and label every decision exit **Yes** or **No**.
5. Trace your pseudocode with the input **13** and fill in one row of **Table 2** for each pass through the loop. The last column should end with 1101.
6. Test your pseudocode and flowchart with the five inputs in **Table 3**. First convert each number by hand, then follow your pseudocode or flowchart step by step. If the two results do not match, find and correct the error in your pseudocode and flowchart.

Part D – Reverse Conversion (Problem 2)

Problem 2: Binary-to-Decimal Converter. Design a program that asks the user for a binary number (entered as digits 0 and 1, such as 1101) and displays its decimal value. Use the **positional (place value) method** from Week 5: starting from the rightmost digit, multiply each digit by its place value (1, 2, 4, 8, 16, …) and add the products.

Worked example (1101 in binary): (1 × 1) + (0 × 2) + (1 × 4) + (1 × 8) = 1 + 0 + 4 + 8 = **13**.

7. In **Box 2**, write the pseudocode. You can take the rightmost digit with **binary MOD 10** and remove it with **binary DIV 10**; the place value starts at 1 and doubles after each digit.
8. In **Figure 2**, draw its flowchart.
9. Test your pseudocode with the inputs in **Table 4**, comparing hand computation with the result of following your pseudocode.
10. **Optional / Bonus:** Extend Problem 1 so the user can choose the target base (2, 8, or 16). For base 16, remainders 10 to 15 must be displayed as the letters A to F. Write the extended pseudocode in the Bonus box.

Part E – Create Your README.md Offline

11. Show file extensions so that you can see the real file name. Press **Windows + E** to open File Explorer, then click **View** → **Show** → **File name extensions** (make sure it is checked).
12. In File Explorer, click **Documents**, then click **New** → **Folder**. Name the folder **cc101-portfolio** (all lowercase, with a hyphen) and press **Enter**. Open it and create two more folders inside: **week07-algorithms** and **images**.
13. Press the **Windows key**, type **Notepad**, and press **Enter**. If your Notepad shows a formatting toolbar, open the **View** menu and switch to the Markdown **syntax** view so the symbols you type stay visible.

14. Type the README content below. Replace everything in square brackets with your own information, and replace the pseudocode with your pseudocode from Box 1. Keep the blank lines; they separate the parts.

```
# CC 101 Computing Portfolio

This repository contains my laboratory outputs and documentation for *CC 101 Introduction to Computing* at **Sorsogon State University – Bulan Campus**.

## About Me

- **Name:** [Your full name]
- **Program and Section:** BSCS 1-[section]
- **Interests:** [two or three interests]
- **Goal:** [one sentence about your goal as a computer science student]

## Course Information

| Item | Details |
| --- | --- |
| Course | CC 101 Introduction to Computing |
| Instructor | [Instructor's name] |
| Academic Year | 2026–2027 |

## Portfolio Contents

1. [Week 7 – Pseudocode and Flowcharts](week07-algorithms/)
2. More activities will be added each week.

## Sample Pseudocode: Decimal to Binary

```
[Paste your pseudocode from Box 1 here]
```

My Workspace

![My laboratory computer](images/my-pc.jpg)

Last updated: [date in YYYY-MM-DD format]
```

15. Save the file. Click **File** → **Save as**. Go to **Documents** → **cc101-portfolio**. Type **README.md** as the **File name**, set **Save as type** to **All files**, make sure **Encoding** is **UTF-8**, and click **Save**.

16. Verify the file. In File Explorer, open the cc101-portfolio folder and confirm the file is named **README.md**, not README.md.txt. If it ends in .txt, right-click it, choose **Rename**, and remove .txt. Then press the **Windows key**, type **cmd**, press **Enter**, and run the commands below. Record the results in **Table 5**.

```
cd %USERPROFILE%\Documents\cc101-portfolio

dir

type README.md
```

Tip: If "The system cannot find the path specified" appears, your Documents folder may be inside OneDrive. Try **cd %USERPROFILE%\OneDrive\Documents\cc101-portfolio** instead.

17. Optional preview: if your Notepad has a formatted view, switch to it from the **View** menu and note in Table 5 which elements appear formatted. Some elements, such as tables, images, and code blocks, may still appear as plain text. GitHub will display them correctly.

18. Copy the whole **cc101-portfolio** folder to your USB flash drive.

Part F – Peer Review

19. Exchange seats with a classmate. Open their README.md in Notepad and complete the checklist in **Table 6** for their file. Write your name as the reviewer and return the worksheet. Fix any missing element in your own file before leaving.

Part G – Take-Home: Upload to GitHub

Do this part when you have Internet access (at home, on campus Wi-Fi, or on your phone's browser in desktop mode).

20. Go to **github.com** and click **Sign up**. Use an email you check regularly and choose a professional username, such as your name. Verify your email address.

21. Click the **+** button at the upper right and choose **New repository**. Type **cc101-portfolio** as the **Repository name**, choose **Public**, and leave **Add a README file** unchecked. Click **Create repository**.

22. On the setup page, click the link **uploading an existing file**. Drag your README.md (and your week07-algorithms folder, if it has files) onto the page.

23. Under **Commit changes**, type **Add initial README** as the message and click **Commit changes**.

24. Open your repository's main page and check that the README appears formatted. Copy the repository URL (for example, https://github.com/username/cc101-portfolio), record it in **Table 7**, and submit it through the LMS.

    Note: GitHub does not store empty folders. The link to week07-algorithms will work after that folder contains at least one file, and the image will appear after you upload images/my-pc.jpg.

V. Data and Findings

Table 1. Problem Analysis for Problem 1 (Part A)

Item	Your Answer
Restated problem (your own words)	Create a program that takes a non-negative whole decimal number, converts it to binary by repeated division by 2, and shows the binary result.
Input	One non-negative integer n (decimal)
Output	Binary equivalent as a string of 0s and 1s
Processes (operations)	Divide by 2 → get remainder; build binary string by placing remainder at front; update number by integer division by 2; repeat
Loop condition (when the loop stops)	Loop while n > 0; stop when n becomes 0
Variables needed	n (input number), binary (string to store result, start empty or "0")

Item	Your Answer
13 and 25 in binary (computed by hand)	13 = **1101**; 25 = **11001**

Box 1. Pseudocode for Problem 1 (Part B)

```
START
   DISPLAY "Enter a non-negative whole number:"
   INPUT n

   IF n = 0 THEN
      SET binary = "0"
   ELSE
      SET binary = ""
      WHILE n > 0 DO
         SET remainder = n MOD 2
         SET binary = remainder & binary
         SET n = n DIV 2
      ENDWHILE
   ENDIF

   DISPLAY "Binary equivalent: " + binary
END
```

Figure 1. Flowchart for Problem 1 (Part C)

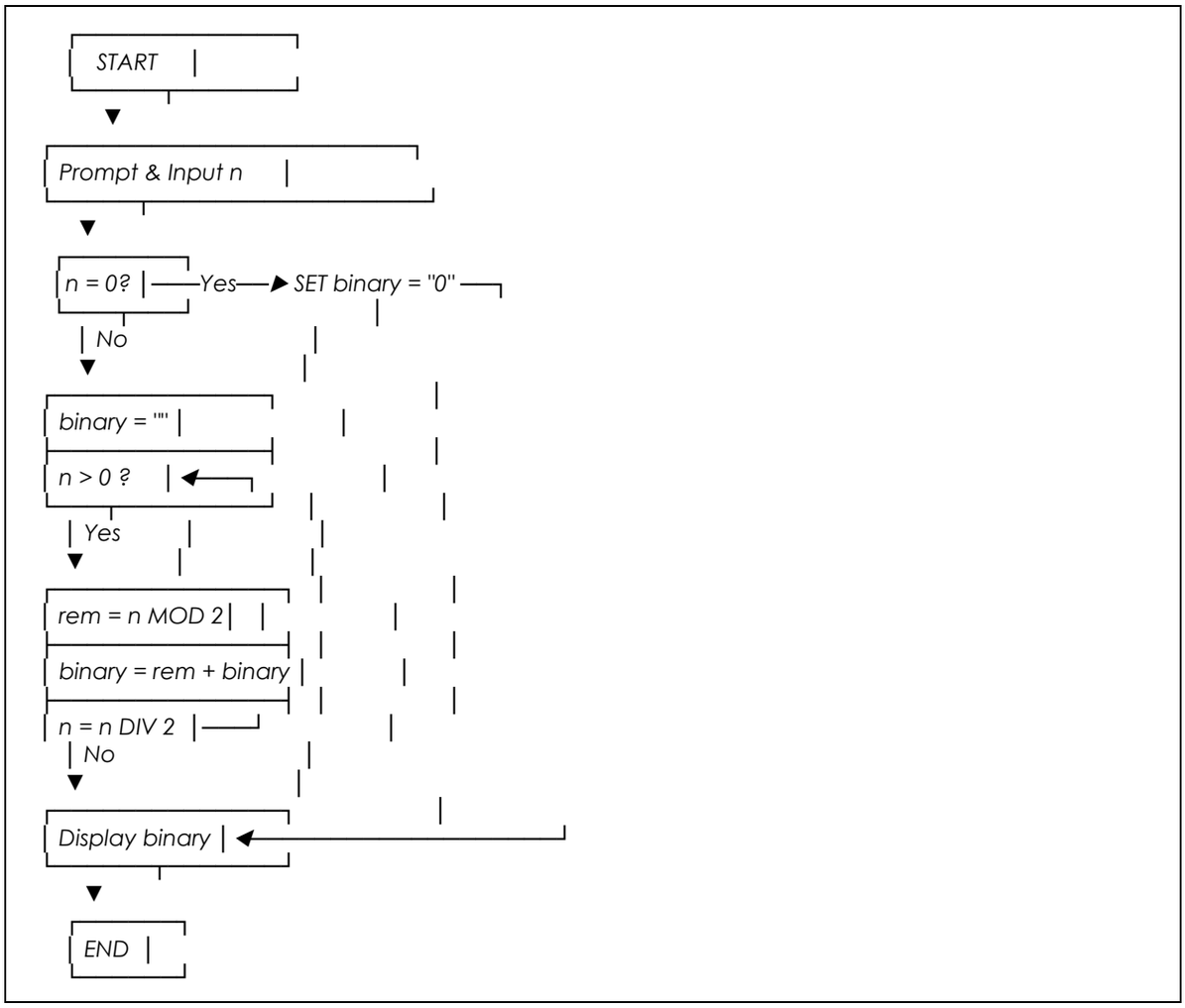


Table 2. Trace of Problem 1 with Input 13 (Part C)

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)
1	13	1	6	"1"
2	6	0	3	"01"
3	3	1	1	"101"
4	1	1	0	"1101"
5	0	(loop ends)	—	—

Value displayed at the end: _1101__

Table 3. Test Cases for Problem 1 (Part C)

Case	Decimal Input	Binary (by hand)	Binary From Your Pseudocode	Match? (Y/N)
1	0	0	0	Y
2	5	101	101	Y
3	13	1101	1101	Y
4	25	11001	11001	Y
5	255	11111111	11111111	Y

Box 2. Pseudocode for Problem 2 (Part D)

```
START
   DISPLAY "Enter a binary number (digits only):"
   INPUT binary_num

   SET decimal = 0
   SET place_value = 1

   WHILE binary_num > 0 DO
      SET digit = binary_num MOD 10
      SET decimal = decimal + (digit × place_value)
      SET binary_num = binary_num DIV 10
      SET place_value = place_value × 2
   ENDWHILE

   DISPLAY "Decimal equivalent: " + decimal
END
```

Figure 2. Flowchart for Problem 2 (Part D)

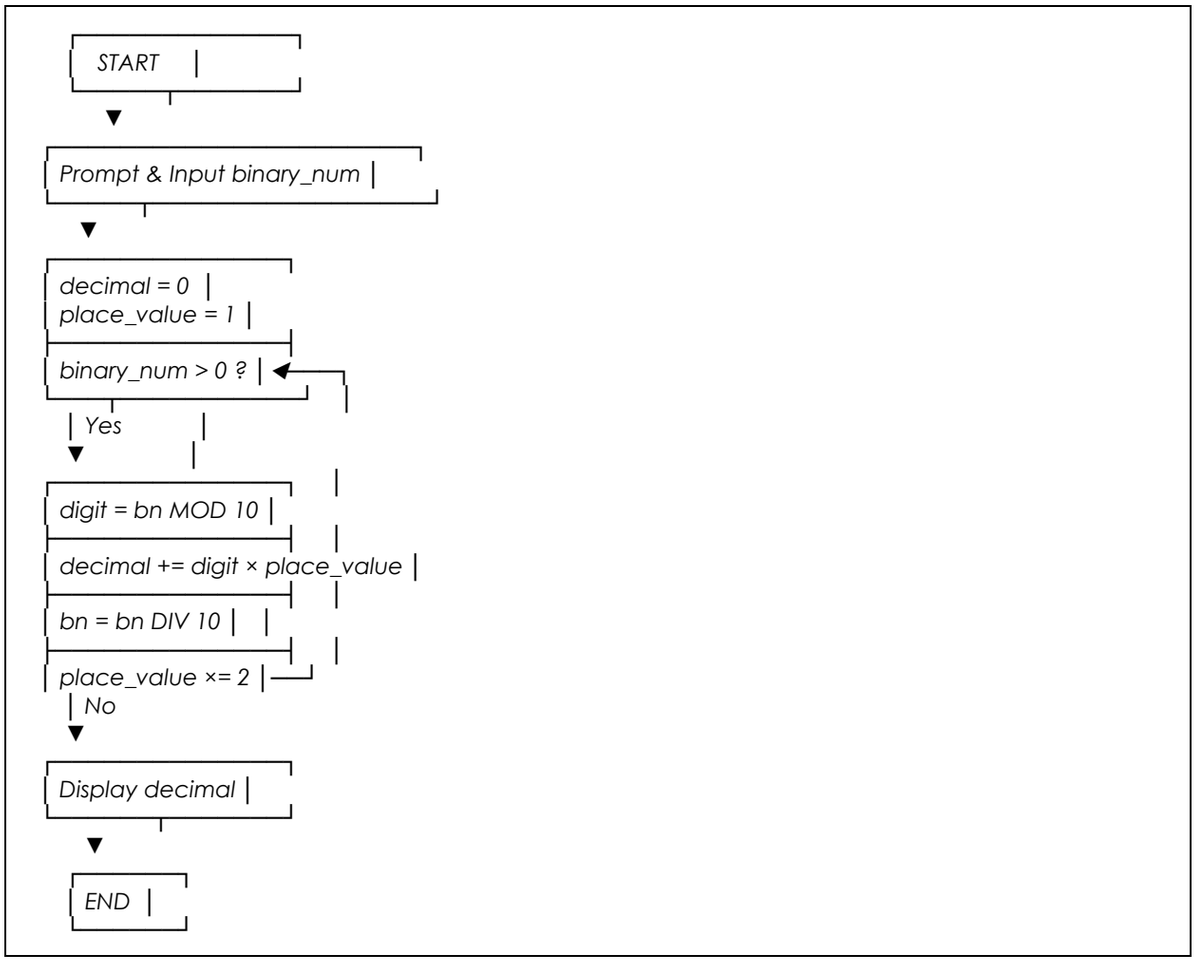


Table 4. Test Cases for Problem 2 (Part D)

Case	Binary Input	Decimal (by hand)	Decimal From Your Pseudocode	Match? (Y/N)
1	1101	13	13	Y
2	101010	42	42	Y
3	11111111	255	255	Y
4	10000000	128	128	Y

Bonus Box. Extended Pseudocode: Decimal to Base 2, 8, or 16 (Optional)

```
START
  DISPLAY "Enter decimal number:"
  INPUT n
  DISPLAY "Enter target base (2, 8, or 16):"
  INPUT base

  IF n = 0 THEN
    SET result = "0"
  ELSE
    SET result = ""
    WHILE n > 0 DO
      SET rem = n MOD base
      IF rem < 10 THEN
        SET digit = STR(rem)
      ELSE
        SET digits = "ABCDEF"
        SET digit = MID(digits, rem - 10, 1)
      ENDIF
      SET result = digit & result
      SET n = n DIV base
    ENDWHILE
  ENDIF

  DISPLAY "Result: " + result
END
```

Table 5. README.md File Check (Part E)

Item	**Your Findings**
Full folder path of cc101-portfolio	"C:\Users\Ali Laceda\Documents\CC101"
File name shown in File Explorer	CC101
Type shown in File Explorer	Markdown Document
Size of cc101 shown by dir (bytes)	~ 1.30 MB (1,372,411 bytes)
Does type CC101 show your Markdown text?	Y
Notepad formatted view available?	N
Elements that appeared formatted in Notepad	None — plain text; GitHub will render formatting

Table 6. Peer Review Checklist (Part F)

Reviewer's name: Francine L. Carilla Owner of the README: Dane Fernandez

Element	**Markdown Used**	**Present and Correct? (Y/N)**
Level-1 heading	#	Y
Level-2 headings	##	Y
Paragraph	Blank lines separate blocks	Y
Bold and italic	**bold** *italic*	Y
Bulleted list	- item	Y
Numbered list	1. item	Y
Table	--- separator	Y
Link	text	Y
Code block	…	Y
Horizontal rule	---	Y
File saved as README.md	—	N

Table 7. GitHub Submission (Part G, Take-Home)

Item	**Your Answer**
GitHub username	**(danefernandez-lab)**
Repository URL	https://github.com/danefernandez-lab/CC101-PORTFOLIO
Date uploaded (YYYY-MM-DD)	October 20, 2026
Does the README display formatted on GitHub? (Y/N)	y
Elements that did not display as expected, and why	

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Which was easier for you to create, the pseudocode or the flowchart? Why?

Pseudocode was easier because it reads like plain English and focuses only on the steps without worrying about drawing symbols and arrows. Flowcharts require more care to place shapes neatly and connect them correctly, but they make the loop logic visually clear.

2. Did tracing and testing reveal any errors in your pseudocode or flowcharts? Describe what you found and how you corrected it.

es — I initially forgot to handle input 0, so it produced a blank result. After testing case 1, I added the IF n = 0 check to set binary to "0". I also fixed the order of concatenation: I had written binary = binary & remainder at first, which reversed the result; swapping it to remainder & binary gave the correct order.

3. How is writing a README in Markdown different from typing a document in Microsoft Word?

Markdown uses plain text with simple symbols like # and * instead of hidden formatting codes. There's no toolbar — you type everything manually. It stays readable even as raw text and works everywhere, whereas Word files depend on the software and store complex binary formatting.

4. How can a GitHub portfolio help you as a future computing professional?

GitHub acts as a permanent, public showcase of my work, code, and growth. Employers often check GitHub to see how I write code, document projects, and follow standards. It also builds a professional presence and makes collaboration easy when I join team projects later.

VII. Scoring Guide

Criteria	Description	Points
Problem Analysis & Pseudocode	Table 1 is complete; the pseudocode for both conversions is correct (correct use of MOD, DIV, and the loop), uses proper keywords, and is indented.	25
Flowcharts & Testing	Both flowcharts use the correct symbols with labeled decisions and a clear loop; the trace (Table 2) and test cases (Tables 3 and 4) are correct.	25
README.md	The file is saved correctly as README.md, uses all required Markdown elements, and passes the peer review in Table 6.	25
GitHub Upload	The README is uploaded to a public cc101-portfolio repository and the URL is submitted (Table 7).	10
Realization / Reflection	Answers are thoughtful, specific to the student's own work, and well written.	10
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	5
Total		**100**